

Sperm production and sperm structure

How do cells in the testis produce and support a specialised gamete?

Learning goal

Distinguish developing sperm, Sertoli cells and interstitial cells, and connect sperm structures with their functions.

Before you begin

Seminiferous tubules produce sperm. The epididymis provides a different location for maturation.

How to complete this lesson Learn → worked example → practise → lesson check

- Read the explanation and its diagrams. Open a bold term when you need a definition.
- Follow the worked example, then try both supported questions. Select Check answer for feedback.
- Open the required check and select Start. Correct both selections to unlock writing.
- Write each explanation and select its Save written response button. Editing saved writing makes it a draft again.
- Select Submit and finish to complete the check. Find your records in All My Work.

Expected work: one completed lesson check. Supported practice, videos and textbook questions are optional. Saved writing is not automatically graded.

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Key terms: spermatogenesis · Sertoli cell · interstitial cell · acrosome

Vocabulary help

Key terms

Words for this lesson

Use these definitions when you need help with a term.

spermatogenesis

The process of producing sperm in seminiferous tubules.

Sertoli cell

A supporting cell within a seminiferous tubule.

interstitial cell

A cell between seminiferous tubules that releases testosterone.

acrosome

An enzyme-containing cap on the sperm head.

More vocabulary for this lesson

Lesson vocabulary**spermatogonium**

A reproductive starting cell that divides by mitosis in the testis.

primary spermatocyte

A reproductive cell that begins meiosis I after DNA replication.

secondary spermatocyte

A cell produced by meiosis I during spermatogenesis.

spermatid

A haploid cell produced by meiosis II during sperm formation.

flagellum

The long tail-like structure that produces sperm movement.

lumen

The inner space of a tube.

differentiation

A process in which a cell becomes specialised in structure and function.

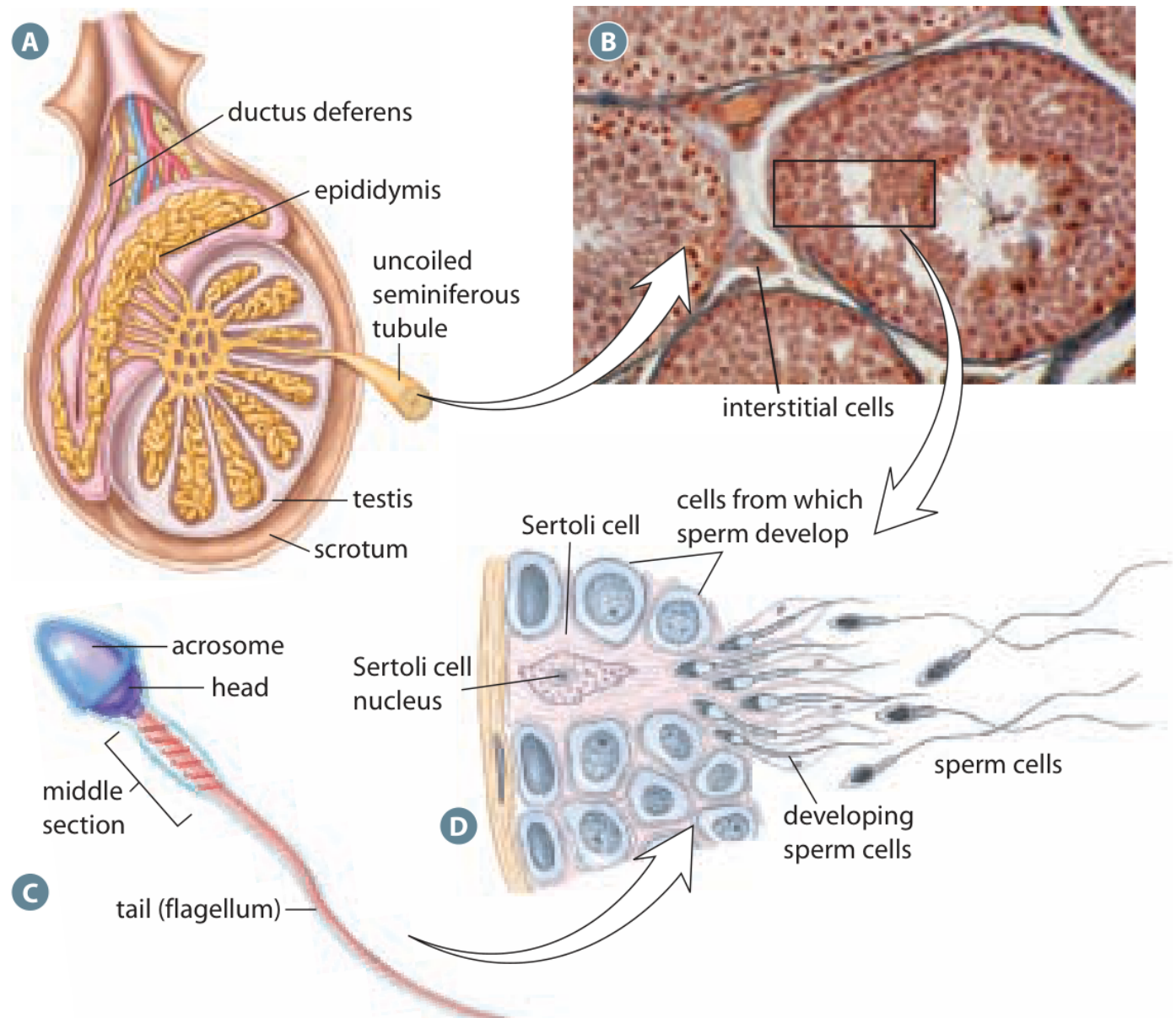
Different cells share the same testis

A testis produces sperm and releases testosterone. That does not mean one kind of cell does both jobs. To explain sperm production, we need to look inside the organ and separate the developing reproductive cells from the cells that support and regulate them.

Spermatogenesis is the process of producing sperm in the seminiferous tubules, the tightly coiled tubes inside a testis. These developing reproductive cells are also called germ cells. They belong to a lineage in which successive stages lead toward sperm. More developed cells are generally closer to the lumen, the inner space of the tubule, where sperm are released.

A Sertoli cell lies within a seminiferous tubule and supports the developing cells. It supplies nourishment and helps maintain their local environment. It remains a supporting cell rather than becoming a sperm. Interstitial cells, also called Leydig cells, lie between the tubules and release testosterone. Inside a tubule and between tubules are different locations with different functions.

In the figure, begin at A with the organ and its coiled tubules. The enlarged view B shows sections through tubules; the interstitial-cell label points to tissue between them. In D, find the tall Sertoli cell beside the smaller developing cells, then follow the developing cells toward the sperm at the right. The broad curved arrows connect enlarged or related views; they do not mean that an interstitial cell turns into a Sertoli cell or a sperm. Panel C shows the specialised cell produced by the reproductive-cell lineage.



Developing sperm and Sertoli cells occupy the tubules; interstitial cells lie between tubules. Inquiry into Biology, p. 480.

Two divisions, followed by a change of form

The starting reproductive cells are called spermatogonia; one is a spermatogonium. They divide by mitosis, a cell-division process that maintains a supply of starting cells. Some descendants develop into primary spermatocytes. Before a primary spermatocyte begins meiosis, its DNA has been copied.

Meiosis is a process with two successive divisions. Meiosis I takes one primary spermatocyte to two secondary spermatocytes. Meiosis II divides each secondary spermatocyte, giving

four spermatids in total. Count the cells at each step: one becomes two, then each of those two becomes two, so $2 \times 2 = 4$. There is no extra DNA-copying stage between meiosis I and meiosis II. After meiosis I, each secondary spermatocyte has one chromosome set, so it is already haploid. Each chromosome still has two attached copied parts, called sister chromatids. These separate during meiosis II without another round of DNA copying.

The four spermatids are haploid: each normally contains one set of 23 chromosomes in humans. Haploid describes the chromosome-set number, not the cell's shape. Meiosis distributes inherited chromosome material into different combinations, so equal chromosome counts do not make the resulting cells genetically identical. The detailed chromosome movements and sources of variation will be examined with meiosis later.

Spermatids then undergo differentiation: they become specialised in structure and function. They develop features such as a compact head and a flagellum. Differentiation is a change of form, not a third meiotic division, and it does not halve the chromosome set again. A group of four spermatids can therefore become four sperm without first producing eight cells.

The sequence to follow is primary spermatocyte, two secondary spermatocytes, four haploid spermatids, then specialised sperm. The teacher slide's phrase 'two cycles of meiosis' should be read as two divisions within one meiosis. Spermatogonia, Sertoli cells and interstitial cells are not alternative names for the four final products.

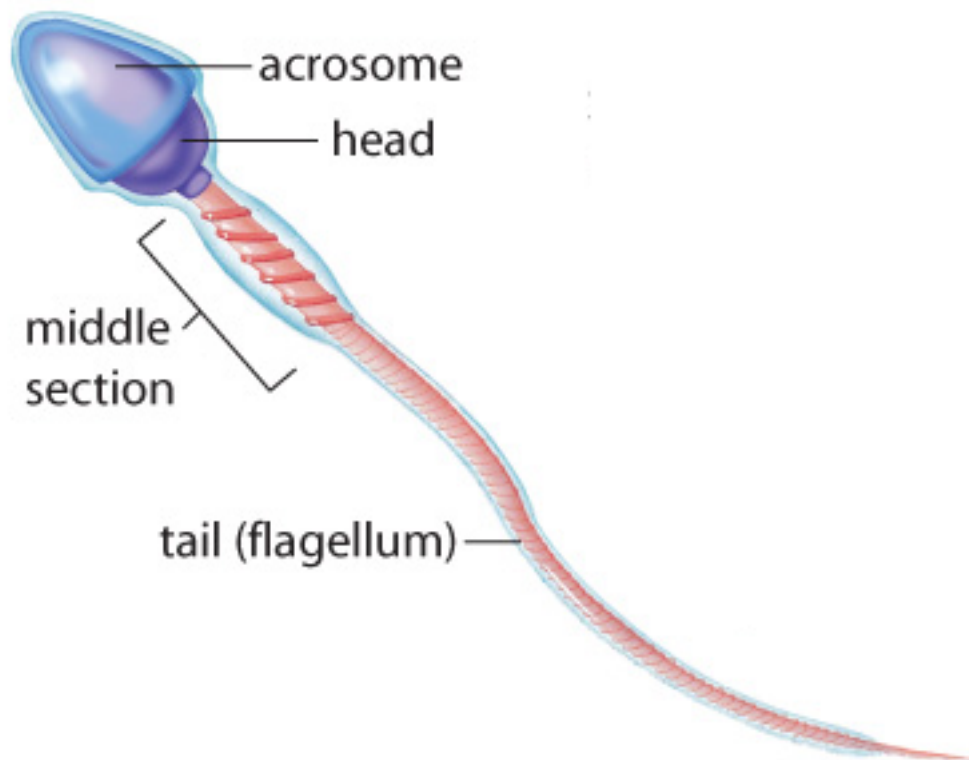
A sperm's parts have different jobs

Look at the sperm figure from its head toward its tail. The head contains a nucleus carrying the haploid chromosome set. The acrosome is the cap at the front of the head. Its enzymes help the sperm pass through the egg's surrounding layers. An enzyme is a molecule that speeds a chemical reaction; these enzymes act on material around the egg.

The middle section, or middle piece, contains many mitochondria. These cell structures help supply ATP, a usable chemical energy source for cell activity. The long flagellum is the tail that produces movement. Its movement requires energy. Supplying energy and producing the movement are connected jobs, but they belong to different structures.

Use the label endpoints in the figure to distinguish the small acrosome cap from the whole head, and the short middle section from the long tail. The illustrated sperm is enlarged and simplified so its parts can be compared. Its drawn length does not give an actual travel distance or speed, and its shape alone cannot establish that each part is functioning normally.

Sperm leaving the seminiferous tubules continue to mature in the epididymis, the coiled duct beside the testis. There they acquire the capacity for movement. Further changes occur in the female reproductive tract before fertilization. These later steps do not change where the cells were originally produced or make the epididymis a site of meiosis.



The acrosome, head, middle piece and flagellum have different roles.

Explain the first affected job

A change in a Sertoli cell first concerns the support given to developing sperm. A change in an interstitial cell concerns testosterone release. A change in an acrosome concerns interaction with the egg's surrounding layers. A change in a flagellum concerns movement. Identify the changed part and explain its normal job before predicting the consequence.

Different changes can lead to a similar observation. Poor movement could involve the movement structure or its energy supply. That observation alone does not identify which component failed. Likewise, a normal count of cells does not prove that the acrosomes or tails function normally. Structure, cell number and function provide different pieces of evidence.

We can now connect the whole process. Germ cells develop through division and differentiation, Sertoli cells support them within the tubules, and interstitial cells supply a hormone from between the tubules. The next question is how signals from the brain and pituitary coordinate those cell roles.

Worked example

A movement problem with a different starting point

A stated classroom model contains sperm with normal-looking heads, acrosomes and tails. The only changed condition is that their mitochondria supply much less ATP than usual. Predict a likely functional effect, and decide whether the appearance of the tails settles the question.

- Locate the changed component: mitochondria in the middle piece. Their relevant job is to help supply ATP, rather than carry the main nuclear chromosome set or provide acrosomal enzymes.
- Connect the component to the dependent process. Flagellar movement uses energy. Reduced ATP supply can therefore reduce effective movement even when the tail still looks structurally normal.
- Test the claim against the given evidence. A photograph can show that a tail is present. It does not measure ATP supply or prove normal movement, so appearance cannot rule out the stated functional problem.
- Keep the conclusion within the model. We have not changed the cell count, measured chromosome number or tested acrosomal enzymes. Poor movement alone would not justify claims that meiosis produced no cells or that testosterone release stopped.

Optional application

A model reports many developing reproductive cells inside seminiferous tubules, but the supporting Sertoli cells no longer provide their usual nourishment. Interstitial cells still release testosterone normally. Which job is directly disrupted, and why does the normal hormone result not establish normal sperm development?

Answer withheld for review

Practise with support

Apply the explanation

Try both questions. Use the hint to revise your thinking after an incorrect answer.

Which cell is inside a seminiferous tubule and supports developing sperm?

- Secondary spermatocyte
- Sertoli cell
- Interstitial cell

Four haploid spermatids have formed. What must happen next to produce specialised sperm?

- Another reduction of chromosome number
- Formation of four interstitial cells
- Differentiation

Stop and think

Try this before opening the explanation

Would four sperm produced from one primary spermatocyte be four identical copies?

Answer withheld for review

Optional video support

Sperm development

As you watch: Distinguish the reproductive-cell stages from their supporting cells.

Optional video. Internet access is required. The explanation below covers the idea without the video.

Read the explanation

A primary spermatocyte completes two meiotic divisions to produce four haploid spermatids. These differentiate into sperm. Sertoli cells provide support; interstitial cells release testosterone.

Optional video support

Following sperm formation

As you watch: Identify where sperm develop and where maturation continues.

Optional video. Internet access is required. The explanation below covers the idea without the video.

Read the explanation

Developing cells are found within seminiferous tubules. Their later stages approach the lumen. Sperm then enter the epididymis, where maturation develops the capacity for movement.

Open required check · Sperm production and sperm structure

Explain what you have learned

This check counts toward chapter progress. Correct the 2 selections to unlock 2 written explanations. Saving writing records your response; it does not grade its accuracy.

Selection 1

A cell lies between seminiferous tubules and releases testosterone. What is it?

- Interstitial cell
- Spermatogonium
- Sertoli cell
- Spermatid

Selection 2

Which sperm structure is most directly involved in helping penetrate the egg's surrounding layers?

- Flagellum
- Acrosome
- Middle piece
- Nuclear chromosome set

Answer the selections correctly to unlock the writing.

Written explanation 1

Explain why Sertoli cells and interstitial cells are not interchangeable. Include location and function.

[Write your own response separately.]

Maximum 3200 characters. Explain the biology in your own words.

Answer withheld for review

Written explanation 2

Trace the main stages from a primary spermatocyte to sperm. Distinguish meiosis from differentiation.

[Write your own response separately.]

Maximum 3200 characters. Explain the biology in your own words.

Answer withheld for review

← Male reproductive structures and the sperm pathway / Next: Hormonal control of the male reproductive system →